

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 3: Fibre to Fabric

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	3 — Fibre to Fabric
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. This hardest paper layers multiple traps inside one stem: it asks you to reason from a property back to its single underlying cause while a competing property is dangled as a decoy, to run a process forward and detect which one inserted step poisons a later step, to separate necessary from sufficient conditions (heat versus agitation, scarcity versus desirability), to keep keratin/fibroin/cellulose and larva/pupa/moth assignments straight under deliberately scrambled options, and to reject a true-sounding fact paired with a wrong reason. Do not write on this paper — mark answers on your answer sheet.

1. A lab gives four results for one mystery thread: it keeps most of its strength when soaked, chars rather than melts, gives a burnt-hair smell, and mats together after hot agitated washing. A student says all four clues are needed to prove it is wool and not silk. Which single clue actually separates wool from silk here?

- (A) The burnt-hair smell, because only wool smells of burnt hair while silk smells papery
- (B) The matting after hot agitated washing, because both wool and silk are protein fibres that char and smell of burnt hair and resist wetting, but only wool's scaly fibres felt and mat
- (C) The wet strength, because silk loses all strength when wet while wool keeps it
- (D) The charring rather than melting, because silk melts like a synthetic while wool chars

2. Three threads are flamed. Thread 1 melts and pulls into a hard bead with no hair or paper smell; Thread 2 chars with a burnt-hair smell and survives wetting; Thread 3 flares

quickly with a papery smell and leaves grey ash but loses strength when wet. A student labels them, in order, 'synthetic, plant, animal.' How many of the three labels are wrong?

- (A) None are wrong; the labels synthetic, plant, animal match the three threads exactly
- (B) All three are wrong, because melting indicates a plant fibre and grey ash indicates an animal fibre
- (C) Two are wrong: Thread 2 is the animal fibre and Thread 3 is the plant fibre, so only Thread 1 (synthetic) is labelled correctly
- (D) Only one is wrong, because Thread 1 should be animal while Threads 2 and 3 are correctly placed

3. A fibre is known to contain nitrogen in its protein and to come from an animal. From this alone, which everyday behaviour can be predicted with certainty, and which cannot?

- (A) Both the burnt-hair smell and the felting can be predicted, because every animal protein fibre felts
- (B) Neither can be predicted, because protein fibres can smell either of hair or of paper
- (C) Felting can be predicted with certainty but the burnt-hair smell cannot, because only some proteins contain nitrogen
- (D) A burnt-hair smell on flaming can be predicted with certainty; whether it felts in hot water cannot, because felting needs surface scales that wool has but silk lacks even though both are animal proteins

4. A merchant insists his white, glossy, smooth thread that loses strength when wet and flares with a papery smell is 'fine mulberry silk.' Reasoning from every clue, the truthful verdict is:

- (A) It is mislabelled cotton, because losing strength when wet and a papery flare are cellulose (plant) traits, whereas real silk is a protein that keeps wet strength and gives a burnt-hair smell
- (B) It is genuine silk, because silk is white, glossy and smooth and flares with a papery smell
- (C) It is a synthetic, because only synthetics are glossy and lose strength when wet
- (D) It is wool, because wool is the animal fibre that flares with a papery smell

5. Wool, silk and cotton are each set alight and each also soaked in water. A student writes a rule: 'A fibre is animal if and only if it keeps its strength when wet.' Test this rule against the three fibres.

- (A) The rule is perfect, because keeping strength when wet is exactly what makes a fibre an animal fibre
- (B) The rule fails as a strict 'if and only if': wool and silk (animal) keep wet strength and cotton (plant) loses it, which fits, but wet strength alone is not a defining property of being animal, so the burn smell is the surer test of the protein/animal nature

(C) The rule is backwards, because animal fibres lose strength when wet while plant fibres keep it

(D) The rule fails because cotton also keeps its strength when wet, just like wool and silk

6. A bag holds three threads, exactly one of which is silk; the other two are cotton and wool in some order. The only test allowed is the burn-smell test. Can the silk be identified for certain by this single test?

(A) Yes, because silk gives a unique papery smell that neither wool nor cotton shares

(B) Yes, because silk is the only fibre that does not burn at all

(C) No, because all three fibres give the same burnt-hair smell, so none can be separated

(D) No, the burn smell alone cannot pinpoint the silk, because silk and wool both smell of burnt hair, so the smell only separates the cotton (papery) from the two protein fibres, leaving silk and wool indistinguishable

7. A herder compares two cold-climate flocks. Flock A averages 5 kg of fleece per animal but is 60% coarse guard hair; Flock B averages 3 kg per animal but is only 10% guard hair. For making the finest, softest yarn after sorting, which flock yields more usable fine fibre per animal, and why?

(A) Flock A, because 5 kg is more total fleece, and more fleece always means more fine fibre

(B) Flock B, but only because lighter sheep are healthier, not because of the guard-hair proportion

(C) Both give the same usable fine fibre, because sorting recovers all the fibre regardless of guard-hair share

(D) Flock B, because 90% of 3 kg is 2.7 kg of fine undercoat, while only 40% of Flock A's 5 kg is 2 kg of fine fibre, so the lighter, finer-coated flock gives more usable soft wool

8. A breeder selects, generation after generation, only the sheep with the finest, most uniform fleece. A rival predicts the flock will eventually 'produce silk-like fibroin because the wool keeps getting finer.' Identify what selective breeding can and cannot achieve here.

(A) It can eventually produce fibroin, because fine enough keratin gradually changes into silk protein

(B) It can produce silk only if the selected sheep are also fed mulberry leaves like silkworms

(C) It can keep refining the wool toward finer, softer, more uniform keratin, but it can never convert that keratin into fibroin, because fibroin is spun by a silkworm larva and no breeding gives a sheep that organ or that protein

(D) It can do neither, because selective breeding cannot change fleece quality at all

9. Two statements are offered: (i) 'Sheep grow thick fleece because their environment is cold,' and (ii) 'India therefore rears its wool sheep in hill states like Jammu & Kashmir and

Himachal Pradesh.' A student says statement (ii) follows logically from statement (i). Is the reasoning sound?

- (A) Yes, the reasoning is sound: because cold climates promote thick fleece, rearing wool sheep in cool hill regions gives the best fleece, so the hill-state practice follows directly from the cold-fleece link
- (B) No, because (i) is wrong: fleece is grown to shed heat, so sheep should be reared on the hot coast
- (C) No, because (ii) does not follow: fleece thickness has nothing to do with the rearing region
- (D) Yes, but only because hill states have more grass, not because of the cold-fleece relationship

10. Under magnification a sheep hair shows overlapping surface scales and a wavy crimp; a silk filament shows neither. Linking structure to behaviour, which single difference best explains why a wool sweater felts in a hot wash but a silk scarf does not, even though both are animal proteins?

- (A) The crimp on wool: the wavy shape tangles in hot water, while silk's straight shape cannot tangle
- (B) The protein difference: keratin always felts and fibroin never can, regardless of surface structure
- (C) The colour difference: dyed wool felts while undyed silk does not
- (D) The overlapping scales on wool: under heat and agitation they ratchet and interlock so the fibres mat, whereas silk's smooth, scale-free filament has nothing to interlock, so it does not felt

11. A wool mill processes fleece in this order: shearing, sorting, scouring, picking burrs, dyeing, spinning. Exactly one early step is in the wrong place, and the error degrades a step that comes much later. Which is the misplaced step, and which later step does it spoil?

- (A) Dyeing is misplaced before spinning; dyeing un-spun fibre ruins the later scouring
- (B) Picking burrs is misplaced before dyeing; removing burrs early spoils the later sorting
- (C) Sorting is misplaced before scouring; sorting greasy fleece grades the fibres unreliably, which spoils dyeing later because coarse and fine hairs end up mixed and the colour and quality are uneven
- (D) Shearing is misplaced first; the fleece should be scoured on the sheep before shearing

12. A manager wants to know the single deepest reason scouring is the 'gatekeeper' that two later steps depend on. Which explanation correctly names both downstream dependencies and why they share the same prerequisite?

- (A) Scouring is the gatekeeper because it both colours and grades the wool, so sorting and dyeing merely repeat it
- (B) Scouring gates only sorting; dyeing actually works better on greasy wool because grease helps dye cling
- (C) Clean fibre is required both to judge grade during sorting and to take dye evenly during dyeing; since grease blocks accurate grading and even dye uptake alike, both steps depend on scouring having removed the grease first
- (D) Scouring gates only dyeing; sorting is done best on greasy fleece because grease holds the grades together

13. A trainee proposes saving a step by combining scouring and dyeing into one hot bath: wash and colour the fleece together before sorting. Predict the two distinct defects this shortcut introduces.

- (A) There are no defects; washing and dyeing together cleans and colours efficiently with no downside
- (B) The dye sets unevenly because grease is still being washed off as the colour goes on, and dyeing before sorting locks coarse and fine hairs into one coloured batch so they can no longer be cleanly graded for fine yarn
- (C) The only defect is that the wool will smell of burnt hair once the hot dye bath touches the protein
- (D) The wool will turn into a plant fibre because the hot dye bath converts keratin to cellulose

14. A worker argues, 'Since scouring uses hot water and also loosens dirt, it must wash out the plant burrs too, so a separate burr-picking step is wasted effort.' Trace the flaw precisely.

- (A) The worker is right; hot scouring water dissolves the burrs along with the grease
- (B) The worker is right; burrs are made of grease and so wash out during scouring
- (C) The worker is wrong, but only because burr-picking must be done before shearing the sheep
- (D) Scouring removes grease, dust and dirt by washing, but embedded woody burrs are solid plant fragments that water does not dissolve or float away, so they remain caught in the fibre and a separate manual burr-picking step is still needed

15. In the wool chain shearing, scouring, sorting, picking burrs, dyeing, spinning, a manager wants to swap one adjacent pair of steps to suit the factory layout. Which swap would do the LEAST harm to the finished yarn, and why?

- (A) Swap scouring and shearing, because the fleece can be washed equally well on or off the live sheep
- (B) Swap sorting and picking burrs, because both act only on already-scoured clean fibre and neither depends on the other's output, so their order between scouring and dyeing is flexible

(C) Swap dyeing and spinning, because colour takes equally well on loose fibre or on finished yarn

(D) Swap scouring and sorting, because greasy fleece can be graded just as reliably as clean fleece

16. An auditor finds a yarn batch is both prickly with embedded plant fragments and uneven in thickness, mixing coarse and fine hairs, yet it is clean and evenly coloured.

Which two steps were performed and which two were skipped?

(A) Sorting and burr-picking were performed, while scouring and dyeing were skipped, which is why the yarn is prickly and uneven

(B) Scouring was skipped (hence the grease) and shearing was skipped, which is why the yarn is uneven and prickly

(C) Scouring and dyeing were performed (hence clean and evenly coloured), while sorting was skipped (hence the coarse-and-fine unevenness) and burr-picking was skipped (hence the embedded plant fragments)

(D) Dyeing and spinning were skipped, which is why coarse and fine hairs remain mixed in the yarn

17. In wool processing, picking burrs comes after scouring but before dyeing. A trainee asks why it cannot simply be done last, just before spinning. Give the reasoning that fixes its position from both sides.

(A) It must come last because only spun yarn holds still enough for burrs to be removed cleanly

(B) Its position does not matter, because burrs fall out by themselves during spinning

(C) It must come before scouring, because washing afterwards rinses away any burr fragments left behind

(D) It must follow scouring because burrs are far easier to spot and remove on clean fibre, and it should precede dyeing and spinning because picking burrs out of dyed yarn would damage the coloured strand and leave debris in the finished thread

18. A sheep gives 5 kg of raw fleece. Scouring removes 40% of the raw mass as grease and dirt. Sorting then discards one quarter of the scoured wool as coarse guard hair. Of the fine wool that remains, dyeing and spinning lose a further 10% as waste. About how much finished fine yarn results?

(A) About 2.25 kg

(B) About 2.5 kg

(C) About 2.0 kg

(D) About 1.5 kg

19. A six-step wool chart asks which single step changes only how the wool looks, without cleaning it, grading it, removing debris, or twisting it. A student answers 'scouring, because clean wool also looks brighter.' Give the precise correction.

(A) The student is right, because scouring is the step that gives wool its colour

(B) The step is sorting, because arranging the fibres changes the wool's appearance most

- (C) Both scouring and dyeing qualify, so the chart has two correct answers
- (D) The step is dyeing, whose sole purpose is to change appearance by adding colour; scouring's brighter look is just a side effect of cleaning, so it fails the 'without cleaning' condition

20. Shearing is scheduled for spring. Four reasons are offered for the timing. Three are valid biology or animal welfare; one is a pseudo-reason that sounds technical but is false. Identify the false one.

- (A) Shearing in spring lets the new fleece regrow before the next winter's cold
- (B) Removing the fleece in winter would leave the sheep dangerously exposed to cold
- (C) In the warm months the sheep does not need its fleece for warmth, so removal causes no heat-loss harm
- (D) 'Wool sheared in spring is automatically finer than wool sheared in any other season, because spring sunlight thins the fibre' — fibre fineness is set by breed and the undercoat, not by the month of shearing

21. Scouring (wool) and reeling (silk) both use water, yet a student claims 'both use water to add the fibre's colour.' Correct the claim and state each step's real, and opposite-feeling, purpose.

- (A) Neither step dyes the fibre: scouring uses water to wash grease and dirt OUT of raw fleece, while reeling uses hot water to soften the sericin gum so the cocoon's single filament can be unwound; colour is added in a separate dyeing step
- (B) The student is right; scouring colours the wool and reeling colours the silk
- (C) Scouring softens gum to unwind wool, while reeling washes grease from the cocoon
- (D) Both use water only to add the final twist that turns fibre into yarn

22. A care label on a wool blanket reads 'cool wash / do not wring / dry flat / no tumble dryer.' A student says each line guards against a different, unrelated risk. Show instead how all four guard against one underlying mechanism.

- (A) Each line guards a separate risk: cool wash protects colour, no wringing protects shape, drying flat protects size, and no tumble dryer protects against fire
- (B) All four protect the dye, because heat and motion wash out colour but never affect the fibre itself
- (C) All four stop the wool from dissolving, because wool dissolves in hot, moving water
- (D) All four limit either heat or mechanical agitation, the two conditions that make wool's scales interlock and felt: cool wash limits heat, no wringing and no tumble dryer limit agitation, and drying flat avoids the rubbing and stretching that also drive felting and distortion

23. Three identical wool mittens are mistreated: Mitten A in hot water, still, no rubbing; Mitten B in cold water with hard rubbing; Mitten C in hot water with hard rubbing. Rank them from most to least felted, and state the rule.

- (A) C most, then A and B roughly tied and much less, then essentially none extra — because felting needs BOTH heat and agitation together, so only C has both; A has heat but no agitation and B has agitation but little heat, so each alone felts far less than the two combined
- (B) A most, then C, then B, because still hot water penetrates deepest and shrinks wool most
- (C) B most, then C, then A, because rubbing alone felts wool regardless of temperature
- (D) All three felt equally, because any contact with water shrinks wool by the same amount

24. A wool fabric and a cotton fabric of identical weave sit in the same room until both reach room temperature, then are touched. The wool feels warmer. A student says 'the wool must therefore be at a higher temperature.' Diagnose the error and give the correct cause.

- (A) The student is right; sitting in the same room, wool ends up hotter than cotton because it is an animal fibre
- (B) The wool feels warmer because it slowly releases heat it generated by a chemical reaction in the fibre
- (C) Both fabrics are at the same room temperature; wool only feels warmer because its crimped, scaly, air-trapping fibres conduct the hand's heat away slowly, so less body heat is drawn off and the hand registers it as warmer
- (D) The cotton feels cooler because it is secretly a synthetic that absorbs the hand's heat

25. A dry wool jumper is warm; the same jumper soaked through feels cold and clammy; wrung out and dried, it is warm again. What single property, and its change, explains the whole sequence?

- (A) The insulating still air trapped by wool's crimp: dry, the air pockets slow heat loss; soaked, water displaces the air and conducts heat far faster, so the jumper feels cold; dried, the air pockets return and warmth is restored
- (B) The wool's protein content: water turns the keratin into cellulose when wet and back into keratin when dry
- (C) The dye: warmth comes from the dye, which washes loose when wet and re-fixes when dry
- (D) Water is a better insulator than air, so the wet jumper should feel warmer, and the clamminess is only a feeling

26. A child whose wool sweater has already felted and shrunk plans to soak it overnight in fabric softener and cold water to 'relax the fibres back to size.' Predict the outcome and the reason.

- (A) It will return to size, because cold water and softener relax the scales and let the fibres slide apart
- (B) It will return to size only if hot water is used, since heat both shrinks and un-shrinks wool
- (C) It will not return to size, because the sweater has chemically become cotton and cotton cannot stretch
- (D) It will not return to size; felting permanently interlocks the fibre scales, and soaking in softener or cold water does not unlock those scales, so the matting cannot be reversed by soaking

27. Three observations about wool are listed: it feels warm, it mats in a hot wash, and it smells of burnt hair when flamed. A student maps all three to 'trapped air.' Give the correct one-to-one mapping of observation to underlying cause.

- (A) All three come from trapped air, so a smooth scale-free wool would still warm, mat and smell of hair
- (B) Warmth comes from the crimp trapping still air; matting comes from the surface scales interlocking; the burnt-hair smell comes from wool being a nitrogen-rich protein (keratin) — three different causes
- (C) Warmth comes from the scales, matting from the protein chemistry, and the smell from the crimp
- (D) All three come from the dye, so undyed wool shows none of them

28. Two scarves of equal mass are made from the same wool: Scarf X is lofty open knit; Scarf Y is dense felt. A student claims the heavier-feeling, denser Scarf Y must be warmer 'because density blocks heat.' Resolve which is warmer and why mass is a red herring.

- (A) Scarf X is warmer, because warmth depends on trapped still air, not on density; the lofty open knit holds more air, while felting Scarf Y squeezes the air out, and since both have equal mass, mass cannot be the deciding factor
- (B) Scarf Y is warmer, because denser fabric always traps more heat by blocking its escape
- (C) They are equally warm, because equal mass means equal warmth regardless of structure
- (D) Scarf X is warmer, but only because lofty knit weighs less and lighter scarves are always warmer

29. The silk-moth life cycle is egg, larva, pupa, adult moth. A quiz asks for the one stage that BOTH feeds heavily AND performs the silk-spinning, and warns that these might be two different stages. Choose correctly.

- (A) Feeding is the larva but spinning is the pupa, because the pupa forms the cocoon as it rests
- (B) Feeding is the larva but spinning is the adult moth, because the moth wraps silk before it flies

- (C) Feeding is the pupa and spinning is the larva, so they are two different stages
- (D) Both are the same stage — the larva: it feeds voraciously on mulberry leaves to grow and then, at the end of its larval life, spins the silk cocoon as it turns into a pupa

30. A narrator says, 'After the moth emerges and flies off, we collect the cocoon it spun and reel one long unbroken thread from it.' A sericulturist objects on two counts. What are the two errors?

- (A) First, the cocoon is spun by the larva, not the adult moth; second, a moth that has emerged has chewed through the cocoon, breaking the single long thread, so you cannot reel one unbroken strand from an emerged cocoon
- (B) The only error is the order; the moth does spin the cocoon, just after laying eggs, not before flying
- (C) The only error is the reeling; the moth does spin the cocoon, but reeling needs cold water, not the emerged cocoon
- (D) There is no error; the emerged moth's empty cocoon gives the longest unbroken thread

31. A sericulture farmer doubles the mulberry leaves fed to each silkworm, expecting roughly double the silk thread per cocoon. Reasoning from what the leaves are actually used for, what really happens?

- (A) The thread length roughly doubles, because the silk is made directly out of the leaves eaten, so twice the food gives twice the silk
- (B) The thread length does not simply double: the larva eats leaves to grow to its normal size and then spins one cocoon whose filament length is set by the larva itself, so feeding it past what it needs is wasted and does not lengthen the single filament
- (C) The worm spins two cocoons instead of one, because surplus food is stored and later released as a second cocoon
- (D) The worm turns the extra leaves into a wool-like fibre that it spins alongside the silk

32. A farmer must decide the exact day to harvest cocoons. Reasoning from the life cycle, the harvest window opens and closes at definite events. When should harvesting happen, and why is the window narrow?

- (A) As soon as the egg is laid, because the egg already contains the wound silk thread ready to harvest
- (B) Just after the larva finishes spinning its cocoon and before the adult moth emerges; the window opens once the unbroken filament exists and closes the moment the moth chews out and breaks it, so it is narrow because waiting too long ruins the long thread
- (C) Only after the adult moth has emerged, because the empty cocoon is the cleanest source of long thread
- (D) During the heaviest feeding of the larva, because that is when the most silk is being eaten and stored

33. Raw silk straight from a cocoon contains fibroin and sericin. A novice tries to weave cloth from the material left after carefully washing away the fibroin and keeping the sericin. Predict the result and assign each substance its true role.

- (A) Weaving succeeds, because sericin is the strong silk thread and fibroin is the throwaway gum
- (B) Weaving succeeds, because fibroin and sericin are equally good weaving threads
- (C) Weaving fails, because both fibroin and sericin are gums and neither can ever be woven
- (D) Weaving fails, because fibroin is the actual silk filament that becomes cloth while sericin is only the gummy coating meant to be washed off during reeling; discarding the fibroin throws away the very thread

34. A reeler is handed one cocoon and told to recover its full thread length but is given no hot water. Predict the outcome and explain the role hot water normally plays.

- (A) The dry sericin gum still binds the filament tightly, so pulling snaps it into short fragments instead of releasing one long strand; hot water is normally used to soften that gum so the single filament unwinds whole
- (B) It unwinds even faster, because a dry filament slides off more easily than a gummed wet one
- (C) It unwinds perfectly, because hot water is only used later to dye the silk, not to free the thread
- (D) Nothing unwinds at all, because a cocoon is solid gum and contains no thread

35. During reeling several cocoon filaments are joined into one strand. A student offers three reasons: (i) to gain usable strength and thickness, (ii) to blend several colours into white, (iii) to thin an over-thick single filament. Which reasoning is correct, and why are the others wrong?

- (A) Reason (ii) is correct; cocoons come in different colours and several are blended to make white
- (B) Reason (iii) is correct; one filament is too thick, so several are combined to thin it down
- (C) Reason (i) is correct — a single fibroin filament is far too fine and weak to use alone, so several are combined for strength and usable thickness; (ii) is wrong because mulberry silk is already whitish so there is no colour to blend, and (iii) is wrong because a single filament is too thin, not too thick
- (D) All three reasons are correct, because combining filaments achieves strength, whiteness and thinning at once

36. Reeling and weaving are both silk steps, but the question is which one actually turns thread into fabric, and what the other does. Choose the statement correct on BOTH counts.

- (A) Reeling interlaces warp and weft to make fabric, while weaving unwinds the cocoon into thread
- (B) Reeling unwinds and combines cocoon filaments into thread; weaving then interlaces lengthwise warp and crosswise weft threads on a loom to make the fabric, so weaving is the fabric-making step
- (C) Both reeling and weaving only unwind cocoons, so neither makes the fabric
- (D) Weaving boils the cocoon and reeling boils it again, and the heat itself forms the cloth

37. Raw silk still coated in sericin feels stiff and dull, while the same thread after degumming feels soft and lustrous. Which single change explains BOTH the new softness and the new shine?

- (A) Degumming dyes the thread, and the dye is what adds both the softness and the shine
- (B) Removing the stiff sericin gum uncovers the smooth fibroin filament beneath: with the hard coating gone the thread bends softly, and the bare smooth fibroin reflects light to give the lustre
- (C) Degumming chemically converts the fibroin into a softer protein, which also happens to reflect more light
- (D) Degumming forces air pockets inside the filament, and those pockets make it both soft and shiny

38. A chart lists fibre sources: sheep, silk-moth larva, Pashmina goat, and the mulberry plant. Three give an animal fibre directly; one gives no fibre at all. Which is the odd one, and why?

- (A) Sheep is the odd one, because sheep are reared for meat and milk, not for any fibre
- (B) The silk-moth larva is the odd one, because it is the adult moth, not the larva, that makes the silk
- (C) The mulberry plant is the odd one: it gives no fibre itself — it only supplies the leaves that feed the silk-moth larva — whereas the sheep gives wool, the silk-moth larva gives silk, and the Pashmina goat gives fine wool
- (D) The Pashmina goat is the odd one, because goats give only milk and never any fibre

39. A guide says muga silk is golden 'because the cocoons are boiled in a golden dye, and all silk would be golden if we dyed it so.' Correct the two false ideas in one accurate statement linking colour, silkworm and region.

- (A) The guide is right; boiling cocoons in golden dye is exactly how muga gets its colour
- (B) Muga's golden colour is a natural property of its particular Assam silkworm, not the result of any dye; boiling only softens the gum, and most other silk (mulberry) is naturally whitish, so silk is not uniformly golden

(C) Muga is golden because the Assam silkworm eats golden mulberry leaves that pass colour into the silk

(D) All silk really is naturally golden and only turns white after bleaching, so muga is just unbleached silk

40. Arrange this complete silk journey so each step follows its prerequisite, noting that the moth-emergence event must be prevented before reeling: mulberry-leaf feeding, weaving, cocoon spinning, killing the pupa by boiling, reeling.

(A) mulberry-leaf feeding, cocoon spinning, reeling, killing the pupa by boiling, weaving

(B) mulberry-leaf feeding, cocoon spinning, killing the pupa by boiling, reeling, weaving

(C) cocoon spinning, mulberry-leaf feeding, killing the pupa by boiling, reeling, weaving

(D) killing the pupa by boiling, mulberry-leaf feeding, cocoon spinning, reeling, weaving

41. A museum needs ONE display card that correctly states both the fibre type AND the source for two exhibits: 'pure Pashmina shawl' and 'pure muga saree.' Which card is correct on all four points?

(A) Pashmina is a SILK from a Himalayan silkworm; muga is a golden WOOL from an Assam goat

(B) Both are silks: Pashmina is mulberry silk and muga is non-mulberry silk

(C) Pashmina is a fine WOOL from the undercoat of a Himalayan goat; muga is a golden SILK spun by a non-mulberry silkworm in Assam

(D) Both are goat wools: Pashmina is white goat wool and muga is golden goat wool

42. A '100% silk' tie is traced to its biological origin. Which description correctly states what the fibre originally was and which life stage made it?

(A) It began as the long fibroin filament that a silk-moth LARVA spun to build its cocoon, later reeled into thread

(B) It began as the fine undercoat sheared from a Himalayan goat (a wool)

(C) It began as the sericin gum scraped from cocoons and woven directly

(D) It began as the seed hairs picked from the bolls of a flowering plant

43. Classify cotton, wool, silk, jute and nylon by origin (plant / animal / synthetic) and by chemistry (cellulose / protein / man-made). Which statement is correct on BOTH classifications for ALL five?

(A) Cotton, wool and jute are plant cellulose fibres; silk and nylon are animal protein fibres

(B) Wool, silk and nylon are animal protein fibres; cotton and jute are plant cellulose fibres

(C) Cotton and jute are plant cellulose fibres; wool and silk are animal protein fibres; nylon is a man-made synthetic — three categories filled correctly

(D) All five are natural fibres because all ultimately come from the earth

- 44.** A single mulberry cocoon is fully reeled and its thread laid out straight. Which description of the length is correct, and what does that length explain about silk's value?
- (A) Only a few centimetres, since each cocoon is tiny, which is why thousands of cocoons make one thread
 - (B) Roughly hundreds to over a thousand metres of one continuous filament, which is why a single cocoon is so prized and why reeling can recover one long unbroken thread
 - (C) Exactly one metre, the same for every cocoon, which is why silk thread is sold by the metre
 - (D) Shorter than a single human hair, which is the reason many cocoons must be combined
- 45.** An exhibit pairs each animal or plant with the fibre it gives. Which single pairing is correct on BOTH the source and the fibre?
- (A) Sheep gives silk; silk moth gives wool
 - (B) Mulberry plant gives silk; sheep gives cotton
 - (C) Silk-moth larva gives silk; Pashmina goat gives wool
 - (D) Yak gives silk; camel gives cotton
- 46.** Among a shepherd, a raw-fleece sorter, a silk reeler and a shopper buying a finished sweater, who is at highest risk of sorter's disease, and what one fact about how the disease spreads decides it?
- (A) The raw-fleece sorter, because the disease spreads by inhaling anthrax spores in the dust of raw, unwashed fleece, and the sorter handles exactly that material before scouring removes the spores
 - (B) The shopper, because body heat from wearing the sweater reactivates dormant spores in the wool
 - (C) The silk reeler, because boiling cocoons releases anthrax spores into the steam
 - (D) The shepherd, because drinking the flock's milk is the main route of anthrax infection
- 47.** A worker who only ever handles already-scoured, dyed wool develops a lung illness, but investigators find no anthrax spores in the finished wool. Which conclusion is best supported about sorter's disease and this worker?
- (A) This worker's illness is unlikely to be sorter's disease from the wool, because scouring removes the anthrax spores, so finished, scoured wool is essentially spore-free and is not the raw-fleece source of the disease
 - (B) It must be sorter's disease, because all wool workers get anthrax regardless of which stage they handle
 - (C) It must be sorter's disease, because dye on finished wool breeds fresh anthrax spores
 - (D) It cannot be any illness at all, because scoured wool is completely harmless to everyone

48. To protect wool workers from sorter's disease, a unit can afford only ONE piece of personal protective equipment. Reasoning strictly from how the disease enters the body, which item gives the MOST direct protection, and why are the others misdirected?

- (A) Rubber gloves, because the spores enter mainly through small cuts on the hands
- (B) A well-fitted face mask (respirator), because sorter's disease is caught by inhaling anthrax spores carried in the dust of raw fleece, so filtering the air the worker breathes blocks the actual entry route
- (C) Steel-toed boots, because the spores settle on the floor and enter through the feet
- (D) Safety goggles, because the spores infect chiefly through the surface of the eyes

49. Silk comes off the cocoon as one continuous filament hundreds of metres long, while wool comes off the sheep as a mass of short, separate hairs. Reasoning from this difference, why must wool be SPUN (twisted) into yarn while a reeled silk strand needs far less twisting to hold together?

- (A) Wool hairs are much heavier than silk, so they need twisting to stop them falling under their own weight
- (B) Wool's short separate hairs would slide apart if simply laid side by side, so they must be twisted together to grip one another into a continuous yarn, whereas silk is already one long unbroken filament that holds its length without the fibres being locked by twist
- (C) Silk needs no twisting because it is really a plant fibre and plant fibres are naturally sticky
- (D) Both need exactly the same amount of twisting, because every yarn is made by the identical process

50. A poster matches four jobs to occupational risks. Exactly one line is correct. Which is it?

- (A) Silk reeler — anthrax from breathing dye fumes during reeling
- (B) Sweater shopper — anthrax from wearing finished, scoured wool
- (C) Shepherd — anthrax only from drinking the sheep's milk, never from fleece
- (D) Wool sorter — anthrax (sorter's disease) from inhaling spores in raw fleece dust

51. An 80% wool / 20% polyester school sweater is criticised by a parent who says 'the polyester is what keeps the child warm; they should make it 100% polyester for more warmth.' Evaluate this reasoning about what each component contributes.

- (A) The reasoning is wrong: the wool's crimped, air-trapping fibres provide the warmth, while the polyester is added for strength, shape-retention and wrinkle resistance, so more polyester would actually reduce the warmth the wool gives
- (B) The reasoning is right; polyester traps far more air than wool, so 100% polyester would be warmest

- (C) The reasoning is right; polyester is an animal fibre that adds the most warmth
- (D) The reasoning is wrong, but only because 100% polyester would smell of burnt hair when worn

52. A factory must improve THREE things at once: wool softness, worker safety from sorter's disease, and garment shape-retention. Which single combined plan addresses all three by the correct mechanism for each?

- (A) Use a coarse breed and skip sorting to save time, let sorters work unmasked, and rely on the wool's grease to keep the shape
- (B) Feed the sheep mulberry leaves for silkiness, skip scouring to keep the fibres soft, and dye brightly to hide spores
- (C) Rear a fine-wool breed and sort the fleece carefully for softness; disinfect raw fleece and give sorters masks and ventilation for safety; and blend in a synthetic like polyester for shape-retention
- (D) Boil the fleece to kill any pupae for safety, sort after dyeing for softness, and add more wool for shape-retention

53. A consumer rejects traditional mulberry silk on the ethical ground that a living insect is killed, and asks exactly which stage is killed, how, and why standard reeling cannot avoid it. Choose the fully correct account.

- (A) The adult moths are crushed while laying eggs, to control the farm's population
- (B) The eggs are boiled before they hatch, so no larvae are ever produced
- (C) The larvae are starved of mulberry leaves until they spin silk out of hunger
- (D) The pupa inside the cocoon is killed by boiling or steaming the intact cocoon before the moth emerges, because letting the moth chew its way out would break the single long filament that reeling depends on

54. A trader prices Pashmina and vicuña fibre very high, arguing 'each animal yields only a little very fine fibre per year, so it is scarce, and being both scarce and desirable, it commands a high price.' Evaluate the full economic reasoning.

- (A) The reasoning is wrong, because Pashmina and vicuña are mass-produced plant fibres that are never scarce
- (B) The reasoning is wrong, because all animal fibres are priced the same regardless of how much each animal yields
- (C) The reasoning is sound: a fine fibre produced in only small amounts per animal each year is genuinely scarce, and a fibre that is both scarce and desirable does command a higher price
- (D) The reasoning is wrong, because scarcity lowers price, since rare goods are harder to sell

55. A sericulture farm splits its cocoons into Batch S (for silk thread) and Batch B (to breed next season's moths). Which difference in handling is essential, and why can the two batches NOT be treated alike?

- (A) Batch S must be boiled or steamed to kill the pupa before the moth emerges, keeping the filament unbroken for reeling, while Batch B must be left UNBOILED so its moths can emerge, mate and lay eggs; boiling Batch B would destroy the breeding stock, and letting Batch S emerge would break every thread
- (B) Both batches must be boiled, because boiling is needed for breeding moths just as much as for thread
- (C) Batch B must be boiled harder than Batch S, because breeding moths need extra heat before they can hatch
- (D) Neither batch is boiled, because boiling is never a part of silk farming

56. A spool sold as 'pure animal-fibre yarn' is tested: it flares with a papery smell, leaves grey ash, loses much of its strength when wet, and never mats in hot agitated washing. Reasoning from every clue together, the honest conclusion is:

- (A) Every clue points to cotton, a plant cellulose fibre: papery flare, grey ash, wet weakness and no felting are all cotton traits, so the 'animal-fibre' label is false
- (B) It is genuine wool, because wool flares with a papery smell and never mats
- (C) It is genuine silk, because silk leaves grey ash and loses strength when wet
- (D) It is genuine Pashmina, because Pashmina is the one animal fibre that never mats

57. A teacher draws a flow showing that warmth, wet behaviour, burn smell and felting each trace back to either structure or chemistry. For a wool fibre, which mapping correctly assigns the CAUSE TYPE (structure vs chemistry) to each of the four behaviours?

- (A) All four trace to chemistry, because every fibre property comes from the protein it is made of
- (B) All four trace to structure, because the way the fibre is shaped decides everything including the smell
- (C) Warmth and felting are structural (crimp traps air; scales interlock), while the burn smell and the protein nature are chemical (nitrogen-rich keratin) — so two behaviours trace to structure and two to chemistry
- (D) Warmth is chemical and the burn smell is structural, because heat is released by the scales

58. A student summarises: 'Selective breeding improves wool, sericulture produces silk, and both can be pushed to give the other fibre with the right effort.' Identify what is true and what is the overreach.

- (A) Everything is true, because with enough effort a sheep can be bred to spin silk and a silkworm to grow wool
- (B) True that selective breeding improves wool and sericulture produces silk; the overreach is the claim that either can be pushed to yield the other, since wool is keratin from a sheep and silk is fibroin from a silkworm larva and no breeding or rearing turns one animal's fibre into the other's

(C) Nothing is true, because selective breeding does not improve wool and sericulture does not produce silk

(D) Only the silk half is true; selective breeding does not actually improve wool quality

59. An ethics-conscious buyer wants a warm, durable, animal-friendly winter garment and considers three options: traditional boiled-cocoon mulberry silk, pure wool, and a wool-polyester blend. Reasoning from how each is made and what each contributes, which best meets warmth, durability AND the animal-friendliness concern about killing?

(A) The wool-polyester blend: wool is sheared from a living sheep (no killing) and gives warmth, while polyester adds durability — so it meets all three, unlike traditional mulberry silk whose standard reeling kills the pupa

(B) Traditional mulberry silk, because reeling does not harm the insect and silk is the warmest and most durable fibre

(C) Pure wool only, because polyester always makes a garment colder and less durable

(D) Any of the three equally, because warmth, durability and animal welfare are unrelated to how a fibre is made

60. A teacher links each of four end-products to the single processing step that most directly makes it possible: a coloured fleece, a usable strong silk thread, a graded soft-wool batch, and a finished saree. Which mapping is correct on all four?

(A) Coloured fleece comes from scouring; strong silk thread from weaving; graded wool from dyeing; the saree from reeling

(B) Coloured fleece comes from sorting; strong silk thread from spinning; graded wool from scouring; the saree from dyeing

(C) Coloured fleece comes from dyeing; strong silk thread from sorting; graded wool from weaving; the saree from scouring

(D) Coloured fleece comes from dyeing; usable strong silk thread comes from reeling (combining filaments); graded soft-wool batch comes from sorting; the finished saree comes from weaving

Solutions — Science (Class 7), Chapter 3: Fibre to Fabric — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	C	21	A	31	B	41	C	51	A
2	C	12	C	22	D	32	B	42	A	52	C
3	D	13	B	23	A	33	D	43	C	53	D
4	A	14	D	24	C	34	A	44	B	54	C
5	B	15	B	25	A	35	C	45	C	55	A
6	D	16	C	26	D	36	B	46	A	56	A
7	D	17	D	27	B	37	B	47	A	57	C
8	C	18	C	28	A	38	C	48	B	58	B
9	A	19	D	29	D	39	B	49	B	59	A
10	D	20	D	30	A	40	B	50	D	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Wool and silk are both proteins, so they share the burnt-hair smell, the charring (not melting), and reasonable wet behaviour; those three clues cannot tell them apart. Felting and matting come from wool's overlapping surface scales, which silk's smooth filament lacks, so the matting clue is the true discriminator. Silk does not smell papery (that is cellulose), so the burnt-hair smell is shared, not unique. Silk does not lose all strength when wet, so wet strength does not separate them. Silk is a protein that chars, not a synthetic that melts, so charring is also shared.

2. (C) A melting bead with no smell is the synthetic signature, so Thread 1 = synthetic is right. The burnt-hair char with wet resistance is the protein/animal signature, so Thread 2 is the animal fibre, not the plant one the student named. The papery flare with grey ash and wet weakness is the cellulose/plant signature, so Thread 3 is the plant fibre. The student called Thread 2 'plant' and Thread 3 'animal', swapping them, so exactly two labels are wrong. Melting indicates synthetic, not plant, and grey ash indicates plant, not animal,

so 'all three wrong' rests on inverted signatures. Thread 1's bead is clearly synthetic, not animal, so 'only one wrong' is also false.

3. (D) All animal protein fibres contain nitrogen and so give the burnt-hair smell on burning, which is therefore predictable. Felting depends on overlapping scales; wool has them and felts, but silk is a smooth protein that does not felt, so 'animal protein' alone does not predict felting. Hence one behaviour is certain and the other is not. Not every animal protein felts (silk does not), so 'both predictable' is wrong. Animal proteins do not smell papery (that is cellulose), so 'either smell' is false. Protein fibres are nitrogen-rich by definition, so the claim that only some contain nitrogen is incorrect.

4. (A) Gloss and smoothness alone do not identify silk; the diagnostic clues are the burn smell and wet strength. A papery flare and loss of wet strength are cotton (cellulose) traits, while silk is a protein that keeps much of its strength wet and smells of burnt hair, so the thread is cotton, not silk. Silk does not flare papery, so calling it silk ignores the burn test. Synthetics melt into a bead rather than flaring with a papery smell and do not necessarily weaken when wet, so 'synthetic' misreads the clues. Wool is a protein that smells of burnt hair, not papery, so the wool label is wrong too.

5. (B) Wool and silk do keep much of their strength wet while cotton weakens, so the pattern superficially matches; but wet strength is a consequence of these particular fibres, not the definition of an animal fibre, so an 'if and only if' rule overclaims. The defining feature is protein chemistry, best shown by the burnt-hair burn smell. So the rule is not perfect. It is not backwards: animal wool and silk do keep wet strength while plant cotton loses it, so the directions are right even though the rule overstates. Cotton loses, not keeps, its wet strength, so the final option is factually wrong.

6. (D) The burn smell sorts fibres into protein (burnt hair: wool and silk) versus cellulose (papery: cotton). It identifies the cotton but cannot tell wool from silk, since both smell of burnt hair, so the silk cannot be singled out by smell alone. Silk smells of burnt hair, not papery, so it shares its smell with wool. Silk does burn (it is a protein), so 'does not burn at all' is false. Cotton smells papery, not of burnt hair, so the three are not all the same; the cotton can in fact be separated.

7. (D) Sorting discards coarse guard hairs and keeps the fine undercoat. Flock A: 40% fine of 5 kg = 2 kg. Flock B: 90% fine of 3 kg = 2.7 kg. So Flock B yields more usable fine fibre despite less total fleece. More total fleece does not guarantee more fine fibre when most of it is coarse, so Flock A's bulk is misleading. The winning reason is the guard-hair proportion and the arithmetic, not the sheep's health. Sorting removes the coarse hairs rather than recovering them, so the two flocks do not end up equal.

8. (C) Selective breeding shifts a trait within what the animal can already make; here it can make keratin finer and more uniform, but it cannot make a sheep secrete fibroin, which is a different protein produced by the silkworm larva's silk glands. So refinement

yes, conversion to silk no. Keratin does not 'change into' silk protein no matter how fine, so the first distractor is biologically false. Mulberry leaves feed silkworms and cannot give a sheep silk glands, so diet does not unlock fibroin. Breeding clearly does change fleece quality (that is its whole purpose), so 'neither' is wrong.

9. (A) Cold prompts thicker, better fleece, so siting wool sheep in cool hill regions (J&K, Himachal) is the logical consequence; both the premise and the inference hold. Fleece insulates against cold rather than shedding heat, so the hot-coast alternative rests on inverted biology. Fleece thickness is exactly why the cool region is chosen, so 'nothing to do with region' is false. The driving reason is the cold-fleece relationship, not grass abundance, so the last option attaches the wrong reason to a correct verdict.

10. (D) Felting is caused by the surface scales of wool ratcheting together under heat plus agitation; silk lacks those scales, so it cannot felt, which is the decisive structural difference. The crimp gives wool bulk and traps air for warmth, but it is the scales (not mere waviness) that cause matting, so the crimp is a decoy here. It is the scales, not an abstract keratin-versus-fibroin rule, that drive felting, so the protein-only explanation misses the mechanism. Colour and dye have no bearing on felting, so the dye option is wrong.

11. (C) You can only grade fibre reliably after washing, so sorting belongs after scouring; sorting greasy fleece mixes grades, and that uneven, mixed batch then takes dye unevenly and spins into poor yarn, so a misplaced early step degrades a much later one. Dyeing fibre before spinning is normal practice and does not 'ruin scouring' (which is already done), so that option is wrong. Burr-picking before dyeing is the correct order, so it is not misplaced. You cannot scour a fleece while it is still on the sheep, so 'scour before shearing' is impossible; shearing must be first.

12. (C) Grease interferes with two different downstream jobs in the same way: it hides the fibre's true texture and length (blocking sorting) and it repels dye unevenly (blocking dyeing). Removing it once, by scouring, unlocks both, which is why scouring gates them together. Scouring cleans; it does not colour or grade, so sorting and dyeing are not repeats of it. Grease repels even dyeing rather than helping dye cling, so 'gates only dyeing... grease helps dye' is wrong. Greasy fleece cannot be graded reliably, so 'sorting best on greasy fleece' is also false.

13. (B) Dye soaks evenly only into already-clean fibre; doing both at once means grease is still leaving as colour arrives, giving patchy dye. Also, colouring before sorting fixes coarse and fine hairs together in one dyed lot, defeating the later grading needed for fine yarn, so two separate defects arise. The shortcut therefore is not defect-free. A hot dye bath does not generate a burnt-hair smell (that needs burning), so that 'only defect' is invented. Heat and dye cannot transmute keratin into cellulose, so the fibre-change claim is impossible.

14. (D) Burrs are tough bits of plant material physically tangled in the fleece; washing removes greasy, dusty matter but cannot dissolve or rinse out solid burrs, so they survive scouring and must be picked out separately. Hot water does not dissolve woody plant fragments, so 'dissolves the burrs' is false. Burrs are plant matter, not grease, so they are not washed out with the grease. Burr-picking is done on the sheared, scoured fleece, never before shearing (there is no detached fleece yet), so that reason is also wrong even though the verdict 'worker is wrong' is right.

15. (B) Sorting (grading the fibre by quality) and picking burrs (removing plant debris) both act on already-scoured clean fibre, and neither one uses the other's output, so their order relative to each other can be swapped with no harm — that swap is the safe one. Swapping scouring and shearing is impossible, since you cannot wash a fleece still on the live sheep. Swapping dyeing and spinning does change the result (colouring loose fibre versus finished yarn is not equivalent), so it is not harmless. Swapping scouring and sorting is the fatal swap: grading greasy, unwashed fleece is unreliable, so that order spoils later steps.

16. (C) Clean, even colour proves scouring and dyeing were done. Unevenness from mixed coarse and fine hairs is the signature of skipped sorting, and embedded plant fragments are the signature of skipped burr-picking, so those two were omitted. The evidence (clean, evenly coloured) shows scouring and dyeing were not skipped, so the first distractor inverts the facts. The yarn is described as clean, so scouring was not skipped, and shearing must have happened or there would be no yarn at all. Dyeing clearly happened (even colour), and skipping spinning would leave no yarn to inspect, so the last option is impossible.

17. (D) Two constraints fix burr-picking's place: clean (scoured) fibre makes burrs visible, so it follows scouring; and removing burrs after dyeing or spinning would mar the coloured thread and leave debris, so it precedes those. Burrs are not removed more easily from spun yarn (they are locked in), so 'last' is wrong. Burrs do not fall out during spinning; they get bound into the yarn, so 'position does not matter' is false. Doing it before scouring means grease hides the burrs, so that order undermines the whole point.

18. (C) Scouring leaves 60% of 5 kg = 3.0 kg. Sorting discards a quarter, leaving three-quarters of 3.0 = 2.25 kg of fine wool. Dyeing/spinning then lose 10%, leaving 90% of 2.25 = 2.025 kg $\approx$ 2.0 kg. The 2.25 kg figure stops after sorting and forgets the final 10% loss. The 2.5 kg figure mishandles the sorting fraction (using a smaller discard). The 1.5 kg figure over-subtracts, e.g. by taking all percentages off the original 5 kg, which double-counts the losses.

19. (D) Dyeing's only job is appearance (colour), satisfying the 'no cleaning, grading, debris-removal or twisting' constraint, so it is the unique answer. Scouring's purpose is cleaning; any brightening is incidental, so it violates the 'without cleaning' rule and the student's reason is wrong on purpose. Scouring does not add colour, so the student's

premise is also factually false. Sorting grades fibres rather than changing their look in the colour sense, so it is not the appearance step. Because scouring is disqualified, there are not two correct answers.

20. (D) Fineness is a property of the breed and its undercoat, not of the shearing month, and sunlight does not 'thin' fibres, so that technical-sounding reason is the false one. Letting a new fleece regrow before winter is a genuine practical reason. Avoiding winter exposure is a real welfare reason. Removing the fleece in warm months when it is not needed for warmth is also valid. So the only invalid reason is the spring-sunlight fineness claim.

21. (A) Scouring's water removes grease and dirt from fleece; reeling's hot water softens the sericin gum so the silk filament releases and unwinds. Neither adds colour, which is a separate dyeing step, so the student's claim is wrong. The second distractor swaps the two purposes (wool has no gum to soften and reeling is not for washing fleece). The twist that makes yarn is a spinning/throwing step, not scouring or reeling, so 'add the final twist' is false.

22. (D) Felting needs heat plus agitation; cool washing removes heat, while not wringing, drying flat and avoiding the tumble dryer all remove agitation and rubbing, so the four instructions share one purpose: preventing felting/shrinkage. They are not four unrelated risks; the unifying mechanism is felting, so the 'separate risks' option misses the link. They protect the fibre structure, not merely the dye, so the dye-only option is too narrow. Wool felts; it does not dissolve in hot water, so the 'dissolving' explanation is factually wrong.

23. (A) Felting requires heat and agitation acting together; Mitten C alone has both and felts most. Mitten A (heat only) and Mitten B (agitation only) each lack one ingredient, so both felt far less than C. So the rank is C, then A and B much lower. Still hot water does not 'penetrate deepest' to out-felt C, so A-first is wrong. Rubbing alone (cold) felts only weakly because heat is missing, so B-first is wrong. Plain contact with water does not felt wool equally; without both heat and agitation little happens, so 'all equal' is false.

24. (C) Both fabrics reach the same room temperature, so they are not at different temperatures; the warm feeling is about the RATE of heat conduction from the hand. Wool's air-trapping structure conducts slowly, so the hand loses heat slowly and feels warm, which corrects the student's temperature error. Wool does not become hotter just by being animal-derived, so the first distractor repeats the error. Wool generates no heat by a chemical reaction, so the second is false. Cotton is a natural plant fibre, not a synthetic, so the third is wrong on both counts.

25. (A) Wool insulates by trapping still air; water conducts heat much better than air, so soaking displaces the air and the jumper feels cold and clammy, and drying brings the air pockets back, restoring warmth, which explains all three states with one property. Wetting does not turn keratin into cellulose, so the protein-change option is impossible. Dye does

not provide warmth, so the dye explanation is wrong. Water is a far worse insulator than air, so the wet jumper genuinely loses warmth; it is not merely a feeling, so the last option is backwards.

26. (D) Felting locks the scales together essentially permanently; neither cold water nor softener unlocks them, so the sweater stays shrunken. Cold water and softener do not make the interlocked scales slide apart, so the first option is wrong. Heat promotes more felting rather than reversing it, so 'hot water un-shrinks' is false. The sweater is still keratin, just matted; it has not become cotton, so the last option's reason is incorrect even though its 'will not return' verdict is right by accident.

27. (B) Each observation has its own cause: crimp traps air (warmth), scales interlock (matting), and protein/nitrogen chemistry (burnt-hair smell). Trapped air explains only the warmth, not the matting or the smell, so the student's single cause is wrong. The second distractor scrambles the three assignments. Dye explains none of them, since undyed wool is still warm, still mats and still smells of burnt hair, so the dye option is false.

28. (A) Insulation comes from trapped still air; the lofty knit (X) holds more air, while felting (Y) compresses the air out, so X is warmer despite equal mass. Density does not equal warmth; packing fibres tight removes the air pockets, so 'denser is warmer' is wrong. Equal mass does not force equal warmth, because structure (air-trapping) governs insulation, so 'equally warm' is false. The two scarves have equal mass, so 'X weighs less' is factually wrong; weight is not the reason.

29. (D) The larva is the heavy feeder, and the same larva spins the cocoon as it transforms into a pupa, so both functions belong to one stage. The pupa rests inside the already-spun cocoon and does not spin it, so 'pupa spins' is wrong. The adult moth emerges from the finished cocoon and lays eggs; it does not spin, so 'moth spins' is wrong. The pupa does not feed (it is the resting stage), so 'pupa feeds' is also false; the feeder is the larva.

30. (A) Two things are wrong: the larva (not the moth) spins the cocoon, and an emerged moth has bitten through it, cutting the long filament into short pieces so no unbroken thread can be reeled. Saying the moth spins it (just at a different time) keeps a false claim, so the 'only error is order' option is wrong. The moth does not spin the cocoon at all, so the second distractor also preserves a false claim. An emerged cocoon is broken, so it gives short pieces, not the longest unbroken thread, making 'no error' false.

31. (B) The larva eats mulberry leaves to grow to its normal full size; once grown it spins a single cocoon whose filament length is fixed by the larva itself, not by how much it was fed past satiation. So doubling the leaves does not double the thread — extra food beyond what the larva needs is simply wasted, making (B) correct. The silk is built by the larva's silk glands, not made 'directly out of' the leaves, so (A) is wrong. One larva spins one cocoon, never two from surplus food, so (C) is wrong, and a silkworm produces only silk, never a wool-like fibre, so (D) is wrong.

32. (B) The unbroken filament exists only between cocoon completion and moth emergence; harvest must occur in that gap, and it is narrow because an emerging moth severs the thread. So the window is bounded by two life-cycle events. Eggs contain no wound silk, so harvesting at egg-laying yields nothing. An emerged moth has broken the cocoon, so waiting for emergence ruins, not improves, the thread. Heavy feeding is before any cocoon exists, so there is no thread to harvest then.

33. (D) Fibroin is the silk filament woven into cloth; sericin is the gum coating removed during reeling. Keeping only sericin keeps the gum and discards the thread, so weaving fails. The first distractor reverses their roles (sericin is the gum, not the strong thread). They are not equally good threads; only fibroin is woven, so 'both good' is false. Fibroin is a fibre, not a gum, so 'both are gums, neither weavable' misclassifies fibroin.

34. (A) Hot water softens the sericin gum so the filament releases smoothly; without it the gum holds the thread fast and pulling breaks it into pieces, so only fragments come off. Dry, gummed thread does not slide off faster; the gum makes it stick, so 'unwinds faster' is wrong. Hot water in reeling frees the thread, not dyes it, so the dye claim is false. A cocoon is one long fibroin filament under a gum coating, so it does contain thread; 'solid gum, no thread' is incorrect.

35. (C) A single silk filament is extremely fine and weak, so combining several yields a usable, stronger thread, making (i) the real reason. Mulberry silk is naturally whitish, so there is no colour-blending need, ruling out (ii). A single filament is too thin to use, not too thick, so combining cannot be to thin it, ruling out (iii). Since two of the three reasons are false, 'all three correct' is also wrong.

36. (B) Reeling produces thread by unwinding and combining filaments; weaving makes cloth by interlacing warp and weft on a loom, so weaving is the fabric step and reeling is not. The first distractor swaps the two definitions. Weaving does not merely unwind cocoons, so 'both unwind, neither makes fabric' is wrong since weaving clearly makes fabric. Boiling belongs to reeling (to soften gum), not weaving, and heat does not 'form cloth' by itself, so the last option is nonsense.

37. (B) Sericin is a stiff gum that coats the fibroin filament; while it is present it stiffens the thread and its rough coating scatters light, so the thread is dull. Degumming strips that coating away, exposing the smooth fibroin beneath — the bare filament now bends easily (softness) and its smooth surface reflects light evenly (lustre), so one change explains both effects, making (B) correct. The shine comes from the bare fibroin, not from a dye, so (A) is wrong. Degumming removes a coating; it does not transmute fibroin into another protein (C) nor inject air pockets into the filament (D), so both are false.

38. (C) The sheep gives wool, the silk-moth larva gives silk, and the Pashmina goat gives a fine wool from its undercoat — each is an animal that yields a fibre directly. The mulberry plant yields no fibre at all; it only provides the leaves that the silk-moth larva eats, so it is

the odd one, making (C) correct. The sheep does give fibre (wool), so (A) is wrong. It is the larva, not the adult moth, that spins silk, so (B) is wrong, and the Pashmina goat does give fibre (fine wool), so (D) is wrong.

39. (B) Muga's gold is a natural feature of its Assam silkworm; boiling softens gum rather than dyeing, and mulberry silk is naturally whitish, so the colour is intrinsic and not universal to silk. Boiling in dye is not how muga gets its colour, so the guide's claim is false. Leaf colour does not pass through to make golden silk, so the diet explanation is wrong. Most silk is whitish, not golden-until-bleached, so the 'all silk golden' idea is false.

40. (B) The larva first feeds, then spins the cocoon; the cocoon is boiled to kill the pupa (and soften the gum) before the moth can emerge and break the thread, then it is reeled into thread, then woven into cloth. So boiling must precede reeling. Putting reeling before boiling lets the pupa live and risks the moth breaking the thread, so that order is wrong. Feeding must come before cocoon spinning (a fed larva spins), so starting with cocoon spinning is wrong. Killing the pupa cannot come before there is even a cocoon or a fed larva, so starting with boiling is impossible.

41. (C) Pashmina is a fine goat-undercoat wool; muga is a non-mulberry silk from an Assam silkworm, so the correct card names a wool from a goat and a silk from a silkworm. The first distractor swaps both fibre types and sources. Pashmina is wool, not silk, so 'both silks' is wrong. Muga is silk, not goat wool, so 'both goat wools' is wrong.

42. (A) Silk fabric originates as the fibroin filament a silk-moth larva spins into its cocoon, recovered by reeling, so the larva is the maker. A goat's undercoat is Pashmina wool, an animal fibre but not silk, so that origin is wrong. Sericin is the gum removed in reeling, not the woven thread, so the fabric is fibroin, not sericin. Seed hairs from plant bolls are cotton, a plant fibre, not silk.

43. (C) Cotton (seed) and jute (stem) are plant cellulose; wool (sheep) and silk (silkworm) are animal protein; nylon is a petroleum-based synthetic, so all three categories are correct. Wool is animal, not plant, so grouping it with cotton/jute is wrong, and nylon is synthetic, not animal protein. Nylon is not an animal protein fibre, so the second distractor miscategorises it. Nylon is man-made, not natural, so 'all five natural' is false.

44. (B) A cocoon is built from one filament often hundreds to over a thousand metres long, which explains both the cocoon's value and why one long thread can be reeled. A few centimetres wildly underestimates it. The length is not a fixed one metre; it varies and is far longer. The filament is vastly longer than a hair, so 'shorter than a hair' is the opposite of the truth; combining cocoons is for strength, not because each is tiny.

45. (C) The silk-moth larva spins silk and the Pashmina goat gives a fine wool, so both halves are right. Sheep give wool (not silk) and the silk moth gives silk (not wool), so that pairing swaps them. The mulberry plant only feeds the silkworm and sheep give wool not

cotton, so that pairing is doubly wrong. Yaks and camels give wool, not silk or cotton, so the last pairing is wrong on both counts.

46. (A) Sorter's disease is anthrax contracted by inhaling spores in raw fleece dust; the sorter handles that very material, so the sorter is at highest risk. Scouring later removes the spores, so a finished sweater carries essentially none and body heat does not reactivate them, ruling out the shopper. Silk reeling involves boiled cocoons, not anthrax-laden fleece, so the reeler option is wrong. The recognised route is inhalation of raw-fleece spores, not drinking milk, so the shepherd-via-milk option misstates transmission.

47. (A) Sorter's disease comes from spores in raw fleece; scouring removes them, so finished scoured wool is essentially spore-free and an unlikely source, meaning this worker's illness is probably not anthrax from the wool. Not all wool workers get anthrax; the risk is specific to handling raw fleece, so 'all get it regardless' is false. Dye does not breed anthrax spores, so that option is wrong. The claim that scoured wool causes no illness 'at all' overreaches; the point is specifically that it is not the raw-fleece anthrax source, so 'cannot be any illness' is unsupported.

48. (B) Sorter's disease is caught by inhaling anthrax spores carried in the dust of raw fleece, so the protection that matches the route is one that filters the air the worker breathes — a face mask (respirator), making (B) correct. The disease is not contracted chiefly through cuts on the hands, so gloves (A) miss the route; it does not enter through the feet, so boots (C) are misdirected; and it is not an eye-surface infection, so goggles (D) are misdirected too. Only the airway protection blocks the inhalation route.

49. (B) Wool comes off the sheep as short, separate staple hairs; laid side by side they would simply slide apart, so they must be twisted (spun) so the fibres grip one another into a continuous yarn — that is why wool is spun, making (B) correct. Silk, by contrast, is reeled off the cocoon as one long unbroken filament that already has its length, so it needs little twist to stay together. Weight is not why wool needs twisting, so (A) is wrong; silk is an animal protein, not a sticky plant fibre, so (C) is wrong; and the two fibres plainly do not need equal twisting, so (D) is wrong.

50. (D) The wool sorter handles raw, unwashed fleece and inhales its spores, so that line is correct. Silk reelers work with boiled cocoons, and dye fumes do not carry anthrax, so that line is false. A shopper's sweater is scoured and spore-free, so wearing it is not a route. The shepherd's recognised exposure would be to fleece-borne spores, not 'only from milk', so that line misstates the route.

51. (A) In this blend the wool supplies warmth through its air-trapping crimp, while polyester supplies durability and shape-retention; so replacing wool with polyester would lose warmth, making the parent's reasoning wrong. Polyester does not trap more air than wool's crimped fibres, so '100% polyester warmest' is false. Polyester is a synthetic, not an

animal fibre, so that claim is wrong. The verdict 'wrong' is right, but the burnt-hair reason is irrelevant and false, so that option attaches a wrong reason.

52. (C) Softness comes from a fine breed plus careful sorting; safety comes from disinfecting fleece and protecting the airway; shape-retention comes from a synthetic blend like polyester, so this plan uses the correct mechanism for each goal. A coarse breed and skipped sorting hurt softness, unmasked sorters are endangered, and grease does not hold a garment's shape, so that plan fails all three. Mulberry leaves are silkworm food, skipping scouring leaves dirty wool, and bright dye does not hide or kill spores, so that plan fails too. A fleece has no pupa to boil, sorting after dyeing is too late for clean grading, and merely adding wool does not give the wrinkle-resistance a synthetic provides, so the last plan also fails.

53. (D) Standard reeling boils the cocoon while the pupa is still inside, killing it so the moth cannot emerge and cut the continuous thread; this is the ethical objection and the reason it is hard to avoid. The killing is of the pupa, not of egg-laying adults, so 'crushed while laying eggs' misplaces the stage. Eggs must hatch into the silk-spinning larvae, so boiling eggs would defeat the process and is not what happens. Larvae are well-fed (not starved) so they spin good cocoons, so the starvation option contradicts practice.

54. (C) Scarcity combined with desirability raises price; an animal yielding only a little fine fibre yearly makes that fibre rare, so the trader's two-part logic holds. Pashmina (goat) and vicuña (a llama relative) are animal fibres, not mass-produced plant fibres, so the first distractor mis-sources them. Animal fibres vary widely in price by quality and quantity, so 'all priced the same' is false. Scarcity of a desirable good raises, not lowers, its price, so the last option inverts basic economics.

55. (A) Cocoons meant for silk (Batch S) must be boiled or steamed to kill the pupa before the moth emerges, because a moth that chews its way out breaks the single long filament that reeling depends on; boiling also softens the sericin so the thread can be unwound. Cocoons kept for breeding (Batch B) must be left unboiled so the moths can emerge, mate and lay the eggs for next season — boiling them would kill the breeding stock. So the two batches cannot be treated alike, making (A) correct. Boiling both (B), boiling Batch B harder (C), or never boiling at all (D) each contradicts one half of this trade-off.

56. (A) A papery flare, grey ash, wet weakness and no felting are all cellulose (cotton) signatures, so the yarn is cotton and the animal-fibre label is false. Wool smells of burnt hair (not papery) and does mat, so the wool option contradicts two clues. Silk is a protein that smells of burnt hair and keeps wet strength, so the silk option contradicts the clues. Pashmina is a wool that mats in hot agitation, so 'never mats' plus a papery flare does not fit it; all clues converge on cotton.

57. (C) Warmth (crimp trapping air) and felting (scales interlocking) are structural; the burnt-hair smell and the protein/wet-strength behaviour come from keratin's chemistry,

so the four split two-and-two between structure and chemistry. Warmth and felting are not chemical effects of the protein, so 'all chemistry' is wrong. The burn smell comes from the protein's nitrogen, not from the fibre's shape, so 'all structure' is wrong. Warmth is structural and the smell is chemical, the reverse of the last option, so it is also wrong.

58. (B) Selective breeding genuinely refines wool, and sericulture genuinely produces silk, so those halves are true; but a sheep makes keratin and a silkworm larva makes fibroin, and neither method converts one into the other, so the 'push to give the other fibre' claim is the overreach. A sheep cannot be bred to spin silk nor a silkworm to grow wool, so 'everything true' is false. Selective breeding does improve wool and sericulture does make silk, so 'nothing true' is wrong. Selective breeding does improve wool quality, so the last option is also wrong.

59. (A) Wool is sheared from a living animal (no killing) and provides warmth, and polyester adds durability, so the blend meets warmth, durability and the no-killing concern. Traditional mulberry silk's standard reeling boils the pupa alive, so it fails the animal-friendliness concern, and silk is not categorically the warmest, so that option is wrong. Polyester adds durability and does not make wool colder in a blend, so 'pure wool only' rests on a false premise. How a fibre is made directly affects animal welfare (e.g. boiling the pupa), so 'unrelated' is false.

60. (D) Dyeing adds colour (coloured fleece); reeling unwinds and combines filaments into a usable strong thread; sorting grades wool into a soft batch; weaving makes the saree, so all four match correctly. Scouring cleans rather than colours, weaving makes cloth not thread, and dyeing colours rather than grades, so the first distractor misassigns every step. Sorting does not colour, spinning is not reeling, scouring does not grade, and dyeing does not weave a saree, so the second is wrong throughout. Sorting does not make silk thread and weaving does not grade wool, so the third also fails.